

Deyu Rao

August 26, 2026

dyrao@ust.hk • deyurao.github.io

6072 LSK Business Building, Department of Economics

HKUST, Clear Water Bay, Hong Kong SAR

Academic Appointment

Assistant Professor, [Department of Economics](#), Hong Kong University of Science and Technology

HONG KONG SAR, *July 2022 -*

Faculty Associate, [Center for Economic Policy](#)

October 2022 -

[Institute for Emerging Market Studies](#)

September 2024 -

[IAS Center for AI for Scientific Discoveries](#)

January 2026 -

[Center for Economic Growth](#)

May 2026 -

Cross-Campus Faulty Affiliate, Hong Kong University of Science and Technology (Guangzhou)

GUANGZHOU, *October 2023 -*

Education

Ph.D., [Applied Economics and Management](#), Cornell University

ITHACA, 2022

M.Sc., [Applied Economics and Management](#), Cornell University

ITHACA, 2017

B.Econ. with Distinction, Energy Economics, [Renmin University of China](#)

BEIJING, 2015

Visiting Student, [Department of Economics](#), University of Warwick

COVENTRY, 2013

B.A., English Language and Literature, [Renmin University of China](#)

BEIJING, 2015

Research and Teaching Interests

Environmental Economics, Urban Economics, Applied Microeconomics

Research

Publications

Research Papers:

Panle Jia Barwick, Shanjun Li **Deyu Rao**, and Nahim Bin Zahur. [The Healthcare Cost of Air Pollution: Evidence from the World's Largest Payment Network](#). *Review of Economics and Statistics*, 106(1), 2026: 110-128. DOI: 10.1162/rest_a_01430.

Shipei Zeng and **Deyu Rao**. [Random Forests with Economic Roots: Explaining Machine Learning in Hedonic Imputation](#). *Computational Economics*, 63(2), 2024: 1-25. DOI: 10.1007/s10614-024-10798-9.

Book Chapters:

Chapter 1, *Trends in and Causes of Global Freshwater Decline*, and

Chapter 2, *Cascading Effects: The Impact of Continental Drying on People, Food, and the Planet*, in Fan Zhang, Christian Borja-Vega, Hrishikesh Arvind Chandanpurkar, James Famiglietti, Rick Hogeboom, Regassa Namara, Zarif Rasul, Pavel Luengas-Sierra, and **Deyu Rao**. 2025. [Continental Drying: A Threat to Our Common Future](#). *Global Water Monitoring*. Washington, DC: World Bank. DOI: 10.1596/978-1-4648-2269-8.

Working Papers

Deyu Rao. [The Role of Environmental Amenities in the Urban Economy: Evidence from a Spatial General Equilibrium Approach](#). 2023. Under review.

Panle Jia Barwick, Dave Donaldson, Shanjun Li, Yatang Lin, and **Deyu Rao**. [Transportation Networks, Short-term Mobility, and Pollution Exposure: Evidence from High-Speed Rail in China](#). NBER Working Paper No. 30462, 2024. Reject and resubmit, *AER: Insight*.

Deyu Rao, Zeyu Chen, Mengwei Lin, and Lin Yang. [Pollution in the Administrative Calculus: Evidence from Land Allocation in China](#). 2026.

Winner, Best Paper Award, CAERE Annual ConferenceNANJING, May 2026

Finalist, Best Paper Award, Conf. on Asia and Pacific Economies (CAPE)SUZHOU, May 2026

Zeyu Chen, **Deyu Rao**, Lin Yang, Wen Wang, and Yatang Lin. Passenger Travel Costs and the Spatial Organization of Economic Activity.

Deyu Rao, Shanjun Li, Ivan Rudik, and Fan Zhang. Water Woes in a Woven World: The Economic and Trade Impacts of Water Stress. 2024.

Wangyang Lai, Shanjun Li, **Deyu Rao**, Yongfan Zhao, and Fan Zhang. The Impact of Water Resources on Trade under a Changing Climate. 2024.

Work-in-Progress

Deyu Rao, Yuta Suzuki, Sifan Xue, and Yun Liu. Trade War “Winners” Revisited: Environmental Consequences of Manufacturing Relocation.

Deyu Rao and Yiyuan Zhou. Bridging Divides: The Influence of Urban Transportation on Economic Equity.

Yan Zhang, Shiyu Wang, and **Deyu Rao**. Do Local Government Officials Tunnel Public Goods to Corporations in China? Evidence from Subway Land Auctions.

Zhengyuan Wu, **Deyu Rao**, and Shanjun Li. Quantifying the Impact of COVID-19 on Economic Activities Through Air Pollution.

Nahim Bin Zahur and **Deyu Rao**. Traffic Congestion and Consumer Spending.

Fellowships, Grants, and Honors

PI, General Research Fund (GRF), Hong Kong Research Grants CouncilJuly 2026 - June 2029

HK\$546,452, “Climate’s New Toll: The Economic Costs of Extreme Weather on Transportation Systems”,
Co-I: Eric Yongchen Zou

Participant, National Natural Science Foundation of China (NSFC), General ProgramJan 2027 - Dec 2030

RMB 400,000, “How Local Government Capacity Shapes Environmental Performance and Ecological Restoration: A Multi-level Analysis Based on Legislative Decentralization Reform”, 地方环境制度供给能力如何影响生态环境治理绩效：基于立法分权的多层次分析，
PI: Yuanning Liang

PI, General Research Fund (GRF), Hong Kong Research Grants CouncilJuly 2025 - June 2028

HK\$827,500, “Bridging Divides: The Influence of Urban Transport’n on Economic Equity”, Co-I: Yiyuan Zhou

PI, Early Career Scheme (ECS), Hong Kong Research Grants CouncilJan 2024 - Dec 2025

HK\$535,443, “Local Governance and Environmental Regulation: Evidence from China’s Urbanization”

PI, Research Grant from Li & Fung Supply Chain InstituteSep 2025 - Aug 2027

HK\$80,000, Co-I’s: Yuta Suzuki and Sifan Xue

PI, Research Grant from Institute for Emerging Market Studies (IEMS)June 2025 - May 2027

HK\$80,000, Co-I’s: Yuta Suzuki and Sifan Xue

Teaching Experience adj. mean [enrollment]

		2023SP	2024SP	2024FA 2025SU	2026SP	2027SP
ECON 2103	Principles of Microeconomics		4.1 [57]	4.6+ [50]	4.7 [68]	*
ECON 4454 ¹	Green Economy and Sustainability	4.6+ [31]	4.7 [17]	4.7+ [52]	4.4 [45]	*
ECON 5670 ²	Urban Economics	4.9 [11]	5.0+ [46]	5.0+ [30]	4.8 [57]	*

¹Previously ECON 4999X ²Previously ECON 6100Q *Scheduled

+Recognition of Excellent Teaching Performance

Finalist, The Franklin Prize for Teaching Excellence 2025, HKUST SBM

May 2026

Professional Activities * scheduled

Invited Seminars and Conference Presentation:

2026: CUHKSZ HSS, HKUST CEP, Fudan CCES, CAPE XJTLU, CAERE, SJTU ICCDS, SUFE, AMES, PKU CCERSI, HKUST-Fudan-SMU Trade, Jinan GLO, SEU Nanjing Urban, SMUE Wuhan U, Hitotsubashi
2025: Tsinghua SEM, Duke Kunshan, HKUSTGZ
2024: SUFE, AERE Summer, RUSE Nanjing, HKU Beijing & Tsinghua CIDE G
2023: ASSA-AREUEA, HKUST CEP, SJTU Antai, PKU, Jinan IESR
2022: HKUST, HKU, NUS, ICCDS, AsRES-AREUEA, UEA, TWEEDS
2021: Cornell, Renmin U, NTU-CEC Workshop
2017-2020: Cornell (CICER & EMI Research Day, AEP, SEERE, Emerging Markets PhD Research Day), NBER Spring Meeting 2017

Discussant:

2026: CAPE, SEU; 2024: SUFE; 2022: ICCDS, AsRES-AREUEA, UEA

Professional Services

Referee:

Economic Journal, Review of Economics and Statistics, Management Science, Journal of the European Economic Association, Journal of Development Economics, Journal of the Association of Environmental and Resource Economists, Journal of Environmental Economics and Management, Regional Science and Urban Economics, Labour Economics, Journal of Economic Behavior and Organization, Journal of Applied Econometrics, Environmental and Resource Economics, China Economic Review, Journal of Transport Economics and Policy, Journal of Policy Analysis and Management, Journal of Risk and Insurance, European Journal of Finance, Energy Journal

Research Committee, UG Committee, Honor Thesis Reviewer, etc., Dept. of Economics, HKUST 2023 -

Part-time Consultant, World Bank Group 2022 - 2025

Judging Panel Member, [Sustainable Smart Campus](#) (SSC) Student Competition, HKUST 2023

Judging Panel Member, [Hong Kong Economic Policy Challenge](#) (HKEPC) 2025

Seminar Committee, Graduate Student Assembly (GSA) of AEM, Cornell University 2018-2019

Selected Press Coverage

“[World Annual Fresh Water Losses Could Supply 280 Million People](#), World Bank, Nov 2025.

“[High-Speed Rail Enables Chinese to Avoid Unhealthy Environs](#),” NBER Digest, Jan 2023.

“[How Smog Affects Spending in China](#),” The Economist, June 2018.

“The Impact of Air Pollution on Healthcare Spending in China,” etc., in [VoxDev](#) (Apr 2019), [VoxEU](#) (Sept 2018), [VoxChina](#) (Aug 2017), [Cornell Econ News](#) (Aug 2018).

Others: [ChinaFund](#) (Jan 2024) on ESG Standards.

Previous Teaching and Research Experience

[Best Ph.D. Dissertation Award](#), Dyson School of Applied Economics and Management, Cornell University 2023

[Outstanding Graduate Teaching Assistant](#), College of Agriculture and Life Sciences, Cornell University 2021

Graduate Fellowship, Summer Graduate Research Assistantship, Cornell University 2017-2021

Scholarship for Academic Excellence, Sports Excellence, Graduate with Distinction, Renmin U of China 2012-2015

Teaching Assistant

[Econometrics](#), AEM 7100 (PhD level), Prof. Shanjun Li Spring 2020, 2021, 2022

[Investments](#), AEM 4670 (UG level), Prof. Sumudu Watugala Fall 2019, 2020

CICER Summer Camp ([Market Simulation](#), Marketing, Behavioral Economics) 2019-2021

Research Assistant

Prof. Shanjun Li, Cornell University ITHACA, 2016, 2018, 2021

[Cornell Institute for China Economic Research](#) (CICER) ITHACA, 2016

[Cornell Institute for Social and Economic Research](#) (CISER) ITHACA, 2016

Prof. Ping Qin, Renmin University of China BEIJING, 2014, 2015

Other Experience

Data Analyst Intern, J. D. Power and Associates	BEIJING, 2014
Field Surveyor, Chinese Residential Energy Consumption Survey	BEIJING & JINING, 2012-2013
Level II Football Referee, Chinese Football Association	BEIJING, 2012-2015
HKUST MSSS Football Team, Faculty and Staff Football Team	HONG KONG, 2022 -

Software Skills

Programming: STATA, MATLAB, ArcGIS, Python, R, SAS, SPSS, ORACLE, MS SQL
Language: Mandarin Chinese (native), English (fluent), Latin (basic)

Student and RA Placements

London School of Economics and Political Science (PhD), Arizona State University (PhD), WashU in St. Louis (PhD), HKUST Guangzhou (PhD), Georgia Tech (PhD), Imperial College (MSc), U Chicago (MPP), HKMA